

Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer CrCl_3

Juan Gomez Quispe,[†] Caique Campos de Oliveira,[†] Carlos Raul Primo
Sapillado,[‡] C Rojas-Ayala,[‡] and Pedro Alves Da Silva Autreto^{*,†}

[†]*Center of Natural and Human Sciences, Federal University of ABC, Santo André, São
Paulo, Brazil*

[‡]*Facultad de Ciencias Físicas, Universidad Nacional Mayor de San Marcos, 15081 Lima,
Perú*

E-mail: pedro.autreto@ufabc.edu.br

Phone: +55 (19) 98279-4988

Supporting Information Available

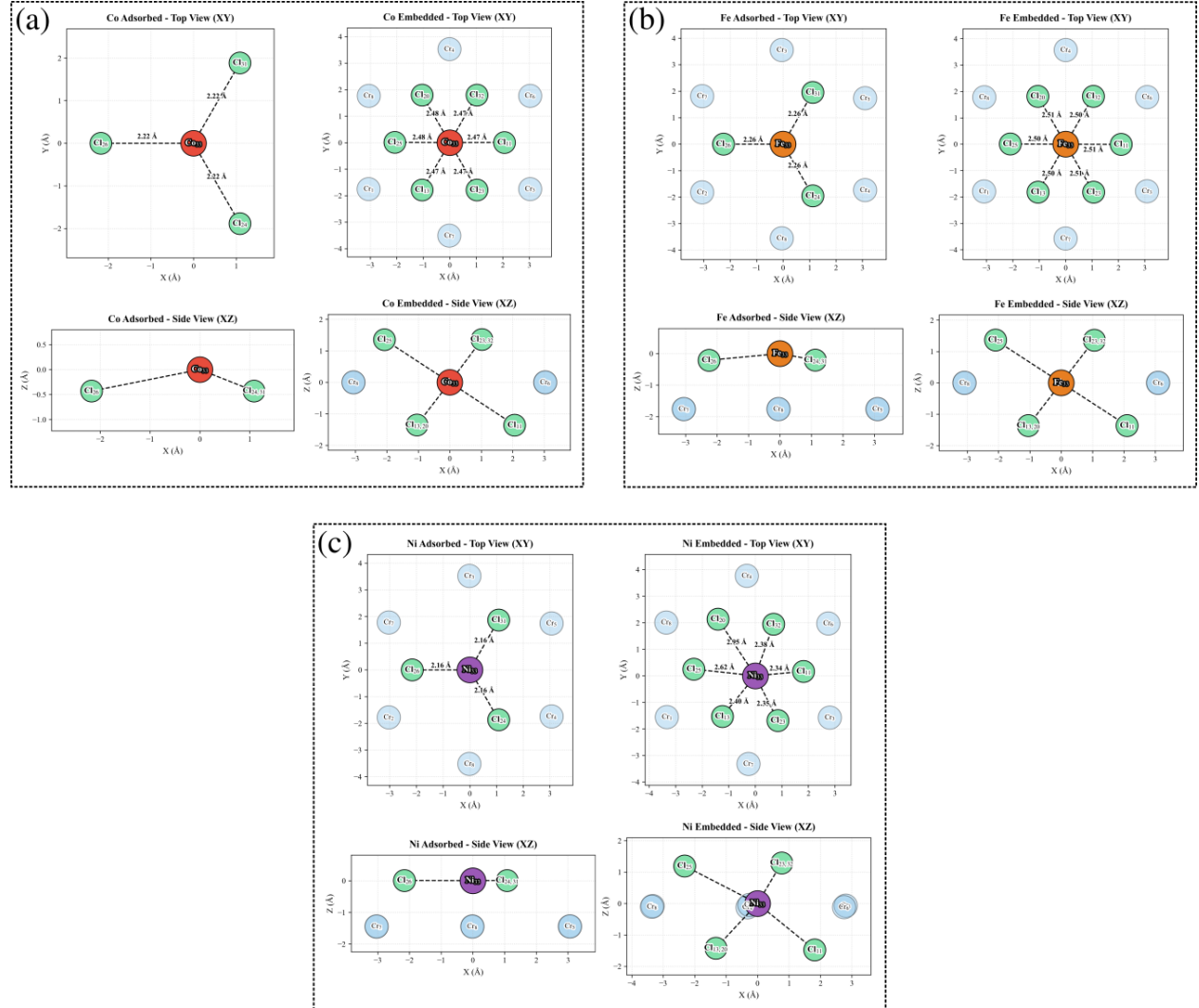


Fig. S1. Schematic top (XY) and side (XZ) views of the optimized local coordination geometries and TM–Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer CrCl_3 in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) configurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

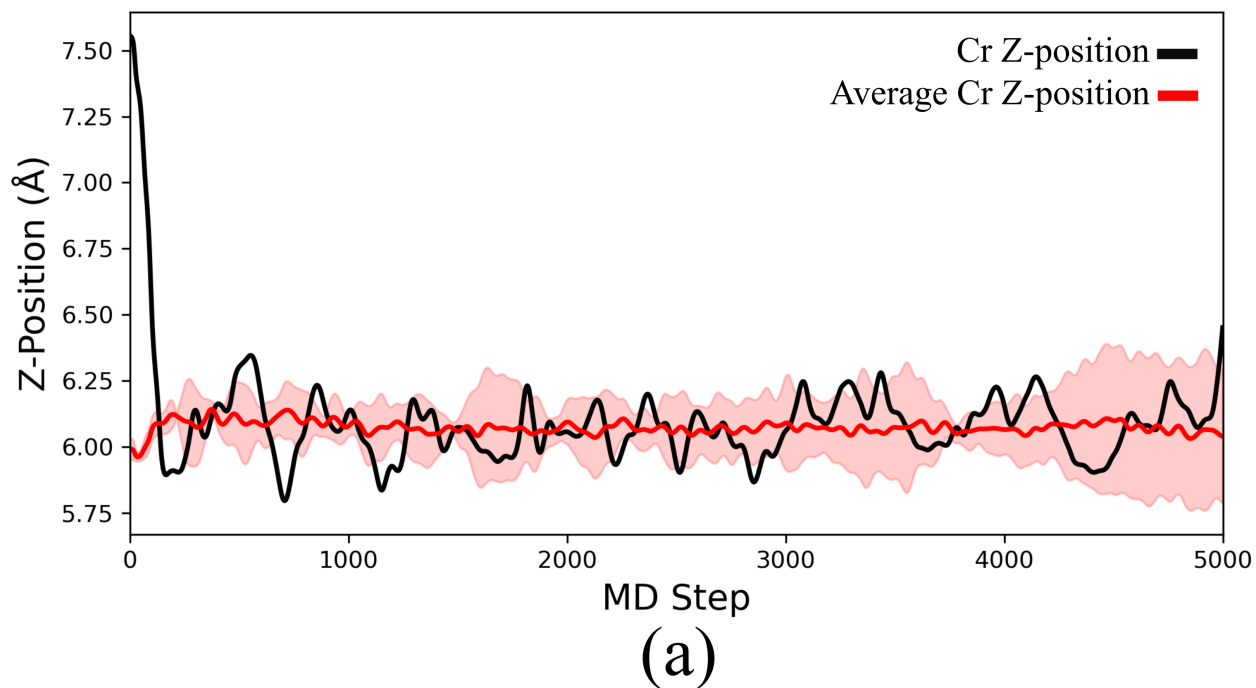


Fig. S2. Time evolution of the absolute z coordinate of the Cr atom initially adsorbed at the hollow site of monolayer CrCl_3 during a 5 ps *ab initio* molecular dynamics simulation at 300 K. The black curve represents the instantaneous z coordinate of the adsorbed Cr atom, whereas the red curve denotes the average z coordinate of the Cr atoms belonging to the host monolayer.

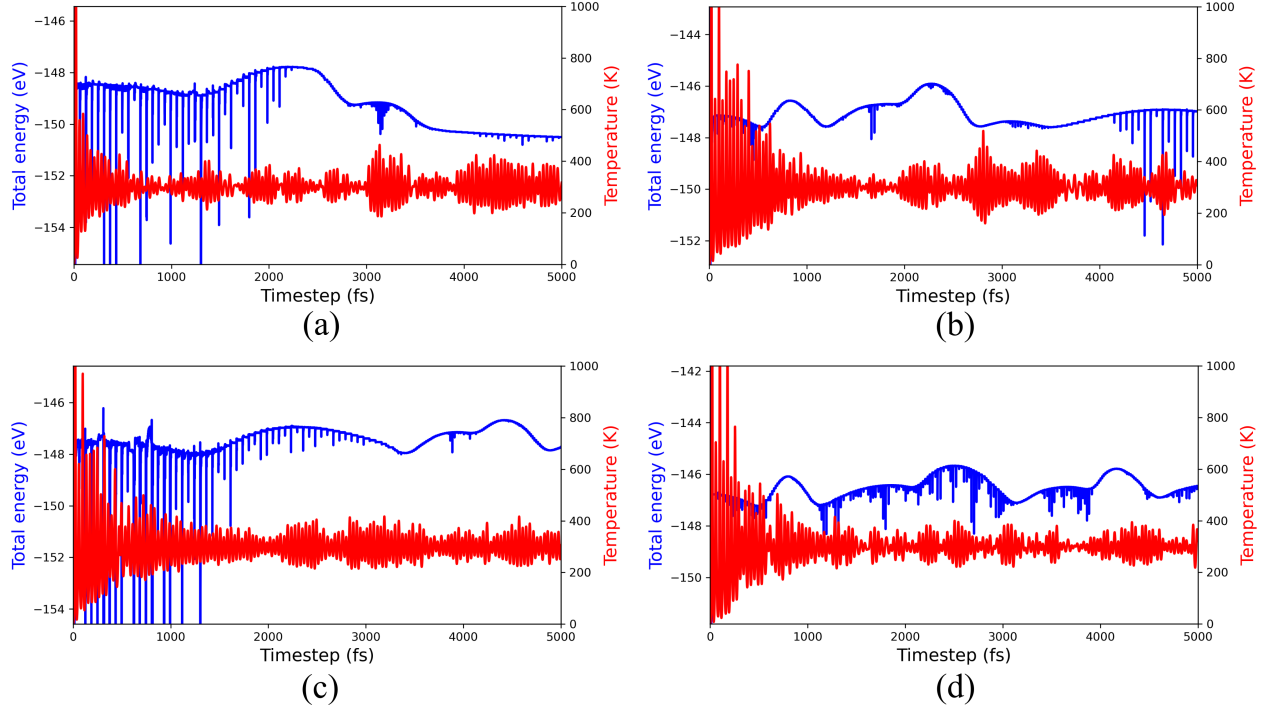


Fig. S3. Time evolution of the total energy (blue curves, left axes) and instantaneous temperature (red curves, right axes) during the 5 ps AIMD simulations at 300 K for monolayer CrCl_3 containing an initially surface-adsorbed (a) Cr, (b) Co, (c) Fe, or (d) Ni atom at the hollow site. Following the initial thermostating transients, the temperatures fluctuate around the target value. The changes in the total-energy profiles accompany the structural rearrangements sampled during the trajectories.

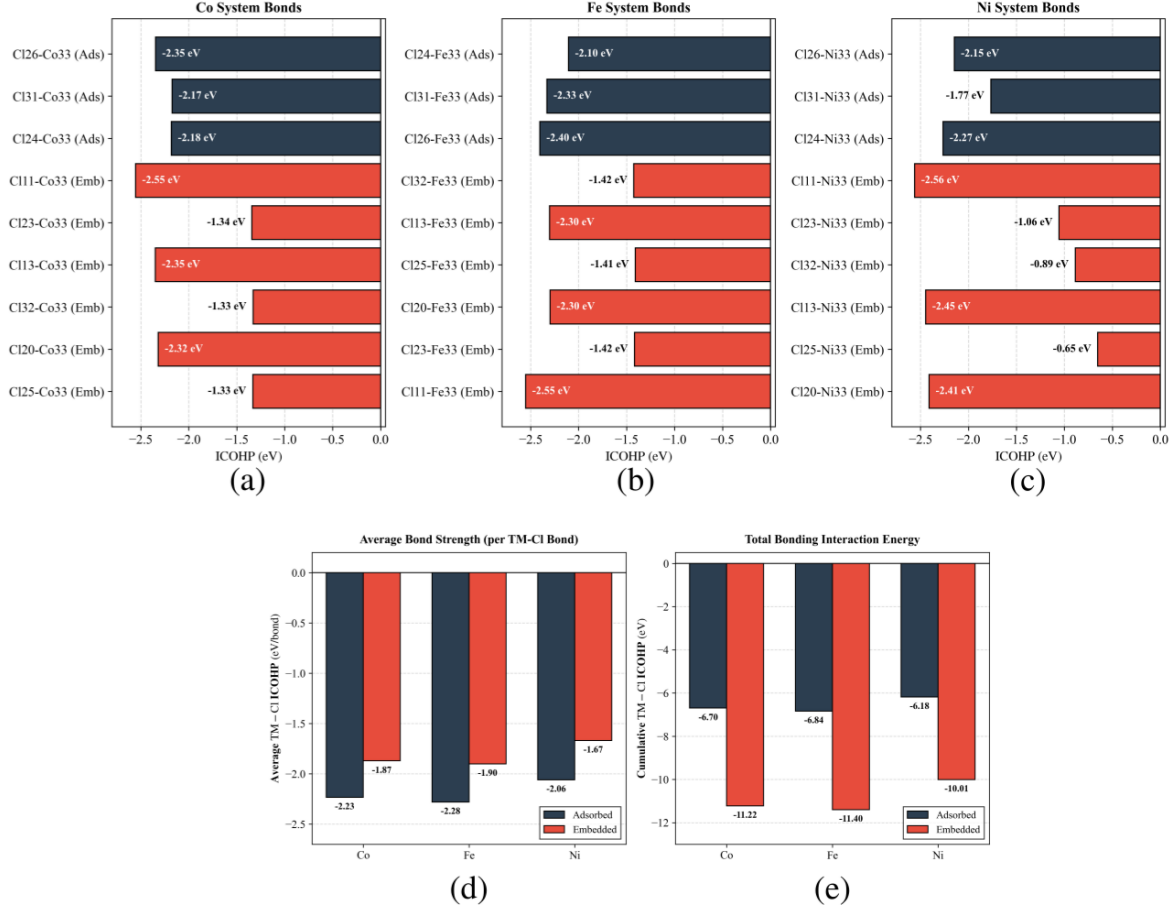


Fig. S4. TM-Cl chemical-bonding analysis and relative stability of surface-adsorbed and embedded transition-metal configurations in monolayer CrCl₃. (a-c) Individual integrated crystal orbital Hamilton population (ICOHP) values for the Co-Cl, Fe-Cl, and Ni-Cl interactions, respectively. Dark bars correspond to the threefold coordinated surface-adsorbed configurations, whereas red bars correspond to the sixfold coordinated embedded configurations. More negative ICOHP values indicate stronger net bonding interactions. (d) Average ICOHP per TM-Cl bond for the adsorbed and embedded configurations. (e) Cumulative TM-Cl ICOHP obtained by summing the individual ICOHP contributions over the corresponding coordination shell.

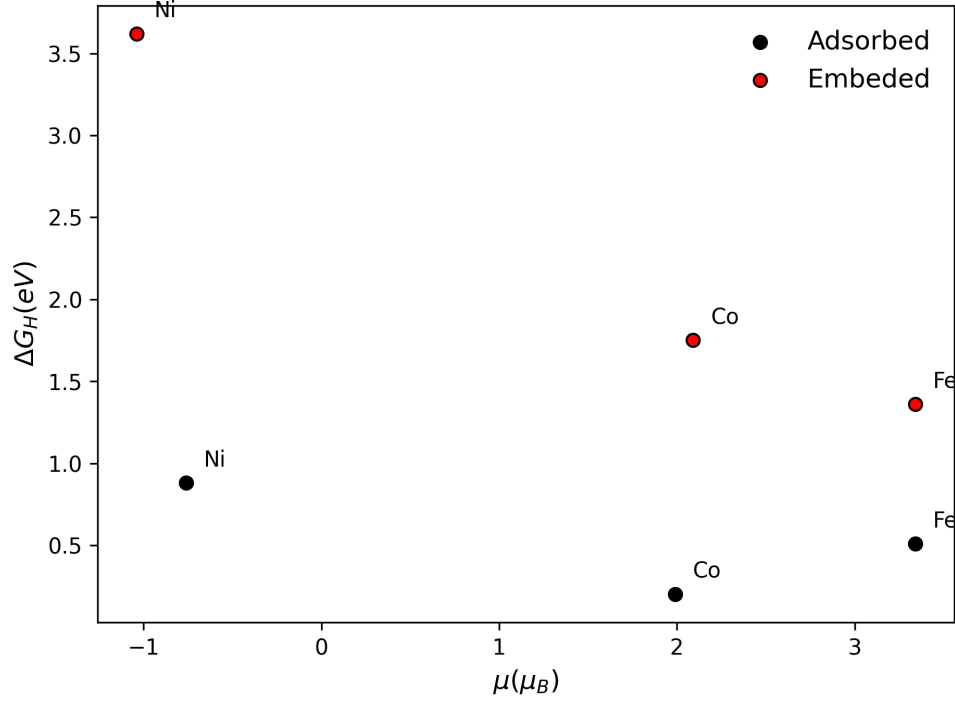


Fig. S5. Relationship between the local magnetic moment of the functionalizing transition-metal atom, μ_{TM} , in the relaxed H-free configurations and the hydrogen adsorption free energy, ΔG_{ads} , subsequently calculated after H adsorption. Black and red symbols represent the surface-adsorbed and embedded Co-, Fe-, and Ni functionalized CrCl_3 monolayers, respectively. Magnetic moments are expressed in units of the Bohr magneton, μ_B .